

CHE 396 Quiz 4

Mike Simon

TOTAL POINTS

98 / 100

QUESTION 1

11 98 / 100

- **0 pts** Correct
- **5 pts** capital cost is wrong (only ISBL is taken into account)
- **3 pts** Gross profit for first year of operation is wrong
- **10 pts** Depreciation is wrong
- ✓ - **2 pts** tax should be next year
- **5 pts** Discount factor is not right
- **0 pts** Click here to replace this description.
- **5 pts** Gross profit is wrong
- **5 pts** tax is wrong
- **10 pts** cash flow is wrong
- **10 pts** taxable income is wrong
- **10 pts** PV is wrong or not done
- **10 pts** NPV is not calculated or wrong
- **2 pts** Depreciation is only taken for ISBL not full capital cost
- **100 pts** Not done or wrong

Michael Simon

Name

OSBL

1. If ISBL cost of a plant is \$60,000 and annual fixed cost of production is \$15,000 and variable cost of production is \$10,000 and annual main product revenue is \$60,000, calculate the maximum 7 year NPV that can be achieved with a 15% interest rate and a 25% tax rate. Assume plant build in two years (capital paid in two years equally), and 50% operation the first year when operation starts. Use MACRS depreciation with a five-year recovery term. (MACRS depreciation charges for 5 year recovery, year 1 = 20, year 2 = 32, year 3 = 19.2, year 4 = 11.52, year 5 = 11.52 and year 6 = 5.76)

to account for OSBL, ICC & working cap
CCP = 1.6 ISBL = 96,000

$$60,000 - 10,000 - 15,000 = 35K$$

$$20,000 - 5,000 - 15,000 = 10K$$

(CF₀)
(111)

YEAR	Gross profit	Depreciation charges	Taxable income	Taxes paid	Cash flow	Discount factor	Present value of CF	NPV
1	0	0	0	0	-48K		-41739	-41739
2	0	0	0	0	-48K		-36294	-78033
3	10K	19200	0	0	10000		6575	-71458
4	35K	30720	4280	1070	33930		19399	-52059
5	35K	18932	15568	4142	30858		15341	-36718
6	35K	11059	23941	5985	29015		12543	-24175
7	35K	11059	23941	5985	29015		10907	-13268
	35K	5529	24471	7367	27633		9033	4245

$NPV_7 = -13268$

$$96,000 \times .2 = 19200$$

$$.32 = 30720$$

$$.192 = 18432$$

$$.1152 = 11059.2$$

$$.1152 = 11059$$

$$.0576 = 5529$$